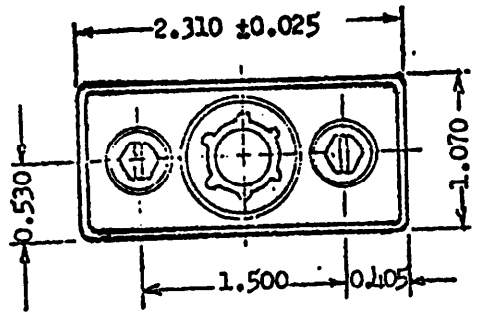
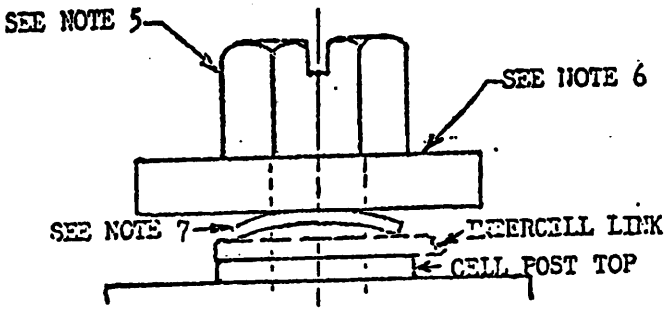
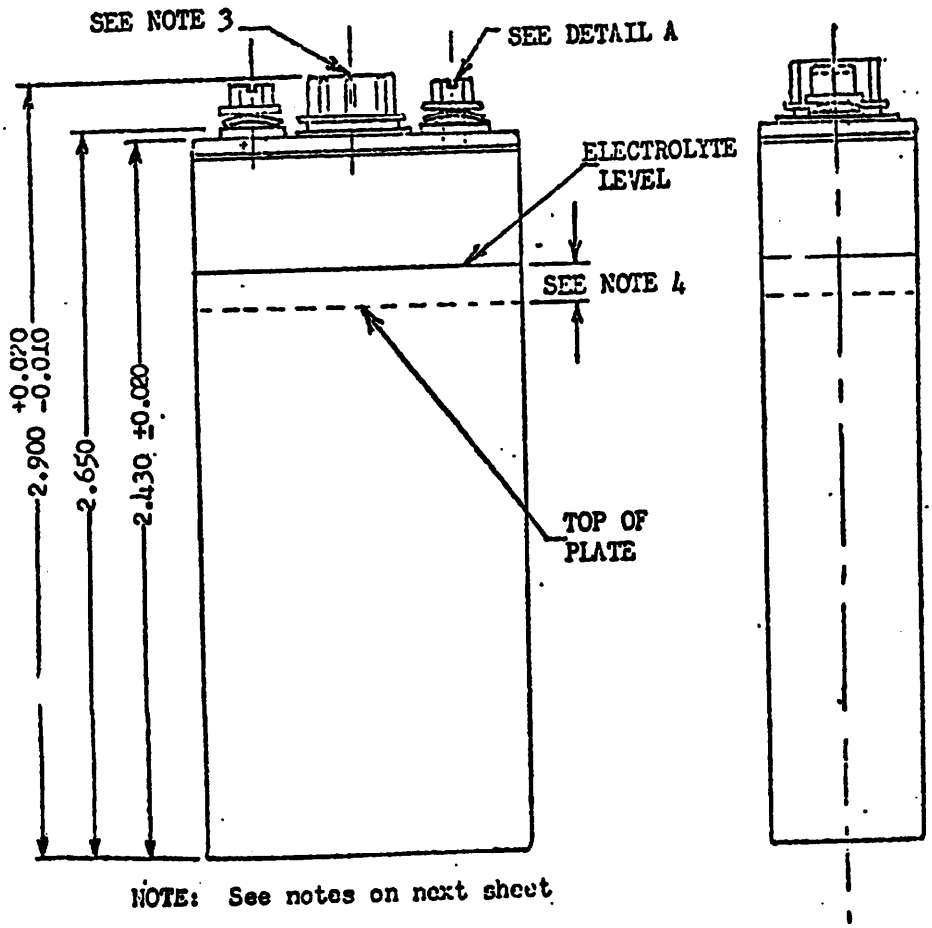
MILITARY SPECIFICATION SHEET
BATTERY, STORAGE BB-616/U

The complete requirements for procuring storage batteries described herein shall consist of this document and the issue in effect of Specification MIL-B-55118(EL).





DETAIL A



INCHES	MM
0.010	0.25
0.015	0.38
0.020	0.51
0.405	10.29
0.530	13.46
1.060	26.92
1.500	38.10
2.310	58.67
2.430	61.72
2.650	67.31
2.900	73.66

NOTE: See notes on next sheet

FSC 6140

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NOTES (Applicable to foregoing drawing):

1. All dimensions shown on figure are in inches. Unless otherwise specified, tolerance is ± 0.015 in.

2. Metric equivalent shown on specification sheet are for general information only and are based upon 1 inch = 25.4 millimeters.

3. The standard bayonet type filler cap shall be Military Part No. MS75046-2.

4. An electrolyte level line should not be furnished on cell's container. The level of electrolyte shall be measurable by a glass tube. Level shall be adjusted with distilled water to 1/4 to 3/8 in. above the top of the plates afte full charge.

5. The slotted hex head screw shall be Military Part No. MS75043-2.

6. The flat washer on each terminal shall be Military Part No. MS75045-2.

7. The Belleville spring on each terminal shall be Military Part No. MS75044-2.

8. The 2.650 height dimension measures from cell bottom to the surface of the cell post top that contacts the bottom of the intercell link, as shown in Detail A.

9. The intercell link shown in Detail A is not furnished as part of the cell when cells are procured.

10. The head-space for the cell is defined as the space between plate tops and bottom of the cell's cover for electrolyte and gas. This space shall be at least twenty (20)% of the overall cell container volume.

11. For design feature purposes, this specification sheet takes precedence over procurement documents referenced herein.

12. Referenced documents shall be of the issue in effect on date of invitation for bid unless a specific exception is taken to same.

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BATTERY, STORAGE
BB-616/U
(CELL)

REQUIREMENTS:

Class: Vented nickel cadmium cell

Dimensions and configuration: See figure and notes.

Weight (max.): 0.53 pounds.

Terminals: Two slotted hex hd screws, each complete with Belleville spring and flat washer per figure and notes.

Container: Polyamide, ABS or equal.

Head space: 20% minimum per Note 10.

Filler cap: Std. per Note 3.

Vent release pressure: 6.0 ± 4.0 psig.

Nominal voltage: 1.2 volts.

Final voltage: 1.0 volt.

Rated capacity, (5 hr. rate): 5.5 amp. hrs.

Normal charge: 0.33 C_5 to 1.50 volts, then 0.17 C_5 for 2.0 hrs.

Normal discharge rate: 0.20 C_5.

0.17 C_5 rate: 0.90 amps.

0.20 C_5 rate: 1.10 amps.

0.23 C_5 rate: 1.27 amps.

0.33 C_5 rate: 1.83 amps.

3.0 C_5 rate: 16.5 amps.

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BATTERY, STORAGE (CELL) BB-616/U

First Article Inspections	MIL-B-55118 Inspection Paragraph	Requirement
Visual and mechanical examination	4.7.1	Table IX
Initial full capacity Discharged (upright)	4.7.14.1	5.5 AH @ 0.20 C_5
Discharged (on side)	4.7.3.1	5.5 AH @ 0.20 C_5
Cycle life	4.7.15	1008 cycles to 3.8 AH
Potassium carbonate	4.7.7	Less than 60 grams per liter
Retention of charge I (14 days)	4.7.13.1	4.9 AH @ 0.20 C_5
Retention of charge II (4 days)	4.7.13.2	2.2 AH @ 0.20 C_5
Low temperature	4.7.14.5	3.3 AH @ 0.20 C_5
Temperature cycling	4.7.10	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	5.5 AH @ 0.20 C_5
Mechanical shock	4.7.11	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	5.5 AH @ 0.20 C_5
Vibration	4.7.12	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	5.5 AH @ 0.20 C_5
Internal pressure	4.7.2	No leakage
Electrical leakage	4.7.9	More than 50 megohms

Quality conformance inspection:

It encompasses same inspections as first article except cycle life shall be 504 cycles to 4.7 AH.

Sampling: See latest issue of MIL-B-55118(EL).

Custodian:
Army-EL

Preparing Activity
Army-EL

Project No. [illegible]

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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